[INCH-POUND]
A-A-2673A
August 10, 1995
SUPERSEDING
A-A-2673
May 1, 1991

COMMERCIAL ITEM DESCRIPTION

NUTRUNNER AND SCREWDRIVER, PNEUMATIC

This Commercial Item Description is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. Scope. This commercial item description covers pistol grip, portable, pneumatic nutrunners and screwdrivers in 2 torque ranges (see Table I). These items are intended for use in installing and removing threaded fasteners.

Table I

National Stock Number	Drive Spindle (in.)	No Load Speed RPM (min. - max.)	Air Consumption CFM (max.)	Torque Range in. - lb. (min.)
5130-00-990-2874	1/4 Hex	750-1350	20	15-50
5130-00-888-6114	1/4 Hex	750-1350	32	25-90

2. Salient Characteristics.

2.1 Design. The nutrunner and screwdriver shall be similar to Figure 1. The drive spindle shall have a 1/4-inch hexagon socket and be capable of completing the specified test and operate properly at air supply pressures between 80-100 psig.

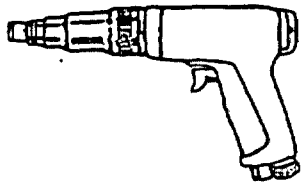


Figure 1. Nutrunner and Screwdriver.

2.2 Motor: The motor shall be of the rotary vane type.

2.3 Bearings: The nutrunner and screwdriver shall be equipped with either ball, roller, or needle bearings or a combination thereof.

2.4 Lubrication: Parts requiring lubrication shall be protected from the entrance of foreign matter and excessive loss of lubricant. The nutrunner and screwdrivers shall provide for the delivery of lubricant in a manner that will assure proper operation.

2.5 Materials: The materials used in the construction of all components shall be sound, of uniform quality and condition, and shall be capable of completing the examinations and tests specified herein. All components shall be new and unused. Repaired or reconditioned components are not acceptable. A metal or composite housing is required. Cadmium plating is not permitted.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center (6FETE-CO), Washington, DC 20406.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:25Z
Check the source to verify that this is the current version before use.

2.6 Handle and throttle: The nutrunner and screwdriver shall have a pistol grip as specified herein. The pistol grip handle shall have a trigger- or button-type throttle. The throttle shall control the volume of air admitted into the motor, automatically shut off the air supply when hand pressure is released, and shall be non-leaking.

2.7 Air inlet connection: The air inlet shall be a well secured, brass or ferrous 1/4-inch or 3/8-inch NPT fitting located at the end of the handle. The air inlet shall be fitted with an air strainer or screen (20 mesh or finer) which shall be effective in preventing solid particles in the compressed air supply from entering the tool. The strainer or screen shall be of such a design that it can be easily removed for cleaning.

2.8 Reversing switch: The nutrunner and screwdriver shall be equipped with a switch that shall change the direction of rotation of the spindle. The switch shall be located so as to be convenient to the operator and shall maintain one position until manually changed.

2.9 Drive spindle: The drive spindle shall have a 1/4-inch hexagon socket as specified. Sockets and tangs shall conform to the requirements of ANSI B107.4. The hexagon socket shall have an externally knurled, quick release mechanism to expedite the use of commercial bits conforming to ANSI B107.4.

2.10 Clutch: The clutch shall be the externally adjustable type and shall be capable of controlling the torque output within the specified ranges and accuracies with the use of not more than two adjustment springs. The clutch shall be of the automatic shut-off type and shall impart no more torque to the driven fastener when the preset torque is reached. The clutch shall automatically reset when the tool is removed from the fastener.

2.11 Manual: One copy of a technical manual consisting of the manufacturer's standard commercial instruction and parts list shall be furnished with each tool.

2.12 Workmanship: Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth, dry, adherent, continuous, and not stained or discolored. Fasteners shall be firmly secured and show no evidence of deformation, cross threading, or hazardous burrs. Adhesives and lubricants adequate for their intended purpose shall be properly and neatly applied. Adhesives shall be adequately cured. There shall be no interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., loose, missing, binding or misaligned parts, sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

3. Regulatory Requirements: In accordance with Section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing supplier's employees to undue hazards from the recovered materials.

3.1 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

4. Quality assurance provisions.

4.1 Responsibility for inspection: Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services adhere to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:25Z
Check the source to verify that this is the current version before use.

4.2 Inspection: It shall be determined that the sample units meet all salient characteristics stated in the requirements section of this document. Any sample containing one or more defects shall be rejected, and if the number of rejected units exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Test conditions: All tests shall be conducted with an air supply of 90 psig (620 kPa) measured as close to the tool as practicable. The test tool shall be connected to the air supply line by means of a 10 foot length of air hose having a hose size of 3/8 inch (inner diameter). The air supply line shall be equipped with a filter, air pressure regulator and lubricator.

4.4 First article inspection. When specified, the contractor shall furnish one sample of a completely fabricated tool for first article inspection. The sample shall have been manufactured and inspected using the same materials, processes, and methods that will be used in the production of the tools used to fulfill the terms of the contract or purchase order. The sample shall meet the requirements of this Commercial Item Description. Approval of the first article sample shall not relieve the contractor of the responsibility to furnish tools in accordance with the requirements of this Commercial Item Description. First article inspection shall consist of the following evaluation in the sequence listed: Inspection, Air consumption test, Speed test, Torque test, Accuracy test, Sound pressure level test, Practical service test, Service test, and Examination after service test.

4.5 Production lot inspection and testing: Production lot testing shall consist of the following evaluations conducted in the sequence listed: Inspection, Air consumption test, Speed test, and Accuracy test. A sample lot shall consist of all items of same NSN offered for delivery at one time. Sampling procedures shall be in accordance with ANSI/ASQC Z1.4 at an Inspection Level of S-3. The acceptable quality level (AQL) shall be 1.5 defects per 100 units tested. A sample unit shall consist of one completely fabricated tool and accessories.

4.6 Air consumption test: Air consumption shall be measured with either a certified liquid displacement meter or a flowmeter.

4.7 Speed test: Each sample tool shall be tested to determine compliance with the speed requirement specified herein. A stroboscope, tachometer or other suitable device, of sufficient capacity and accuracy, shall be used.

Table II

National Stock Number	Torque Settings in. - lb. (± 10%)	Maximum Spread in. - lb.
5130-00-990-2874	15	4
	30	6
	50	8
5130-00-888-6114	25	5
	60	8
	90	10

4.8 Torque test: Each sample tool shall be tested to determine compliance with the torque requirement specified herein. The test shall be conducted on a prony brake analyzer or similar system of appropriate capacity calibrated to assure accurate readings.

4.9 Accuracy test: Each sample unit shall be tested on a dynamic torque analyzer that conforms to ISO 5393 for repetitive torque setting accuracy at the applicable torque values specified in Table II. Accuracy shall be tested with the tool operating in the clockwise (right hand) direction only. Ten readings shall be made at each torque setting and the maximum spread in torque readings shall not exceed the values specified in Table II.

4.10 Sound pressure level test: The sound pressure level test shall be conducted in accordance with the applicable provisions of the Compressed Air and Gas Institute's, "CAGI-PNEUROP Test Code for the Measurement of Sound from Pneumatic Equipment". The sound pressure level shall not exceed 90 decibels on the A weighted scale.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:25Z
Check the source to verify that this is the current version before use.

4.11 Practical service test: Each sample tool shall be operated for 5 minutes. The test shall duplicate a typical work application. During the test, the tool shall be checked for throttle operation, rotation, excessive noise or vibration, ease of operation, and general serviceability.

4.12 Service test: Each sample nutrunner and screwdriver shall be subjected to the following service test. The tool shall be operated manually or installed in a fixture in a manner that the tool can be loaded and automatically actuated for 1000 cycles. Each cycle shall be a minimum of 5 seconds ON and a maximum of 5 seconds OFF. The load applied shall be the high limit of the specified torque range for the applicable tool. Any tool that fails to complete the 1000 cycles will be rejected.

4.13 Examination after service test: Tools satisfactorily completing the service test shall be subjected to the Sound pressure level test, Air consumption test, and Accuracy test, and shall conform to the applicable requirements of these tests. Upon completion of the Service test, the tool shall be disassembled and the parts examined. Any tool having broken, cracked, or deformed parts shall be rejected.

5. Preservation, packaging, packing, and marking: Tools shall be preserved, packaged, packed, and marked as specified in the contract or purchase order.

5.1 Identification markings: Each item (including all items within a set or kit) shall be marked with the manufacturer's name or identifying symbol, and the state or country of manufacture, unless otherwise specified. All marking shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

6. Notes:

6.1 The issue of referenced documents in effect on the date of invitation for bids or request for proposal shall be used to determine compliance with the stated requirements. Sources for obtaining referenced documents may be determined by contacting the Contracting Officer.

6.2 Military standards are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19120-5094.

6.3 ANSI standards are available from the American National Standards Institute, 1430 Broadway, New York, NY 10018.

6.4 The "CAGI-PNEUROP Test Code for the Measurement of Sound from Pneumatic Equipment" is available from the Compressed Air and Gas Institute, 1300 Sumner Avenue, Cleveland, OH 44115.

Military Coordinating Activity

Army - GL

Preparing Activity

GSA - FSS

Custodians

Army - GL
Air Force - 99

Review Activities

Air Force - 35, 84

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:25Z
Check the source to verify that this is the current version before use.